Poor information also impairs liquidity, and liquidity in muni markets is abysmal. Transaction costs are high, price adjustment is slow, and different buyers pay different prices for the same bond. The bonds themselves are overly complex, and this further reduces liquidity. The way muni bonds are issued in series is ironically designed to minimize liquidity: it takes a large issue and then chops it up into several small issues (thirteen, on average, but with 5% of series issues containing more than twenty-five separate securities). Each small issue has a different maturity, different coupon, and perhaps even different embedded derivatives. There are literally a million or so different bonds in the muni market. Over 60% of them contain hard-to-value embedded derivatives. Contrast this with the large, deep U.S. Treasury market. U.S. Treasuries, unlike munis, are simple and easy to value, and there are relatively few issues with large sizes.

As a result of all this, individual investors often get fleeced when buying munis. Investors need to pay attention to recent comparable bond prices, released through the Municipal Securities Rule Marking Board, but the vast array of bond types and features makes these securities hard to compare; in addition to different maturities, coupons, and discounts/premiums, munis can be sinkable, callable, putable, refunded, conduit, bundled with other contingent claims, subject to alternative minimum tax, taxable, bank-qualified, and so on.

Round-trip transaction costs for individuals average 2% to 3% but can easily reach 5%. This is more than double what institutions pay, roughly twice as much as what it costs to trade a corporate bond, and many times what it costs to trade a stock. Green, Hollifield, and Schürhoff (2007) show that dealer markups over the reoffering price can be high as 5%—roughly a year's return on a typical muni bond. (The reoffering price is often represented to issuers as the price at which the bonds are sold to the public.) Biais and Green (2005) show that, remarkably, it is twice is as expensive to trade munis today as it was in the 1920s, when muni bonds were actively traded on the New York Stock Exchange. Given the huge advances in technology, what other securities cost more to trade today than they did 100 years ago?

Figure 12.2 shows just how illiquid muni markets are. Dividing up all municipal bond transactions into ten buckets by the number of trades per year, excluding the bonds that never trade and taking bonds transactions which occur ninety days after issue, the 10% most illiquid bonds trade once every five to six years. The typical municipal bond trades once or twice per year. Even the most liquid bonds trade once every two days, on average. Muni bonds are extremely illiquid.

It's tragic that these bonds, which appeal mainly to individual investors, are so hideously inefficient and unfriendly. Reforming this market would immensely benefit the common man as well as save billions in taxpayer dollars by lowering issuer costs. But muni reform is hard: the Securities and Exchange Commission (SEC) does not regulate this market and cannot impose mandatory disclosure requirements or minimal accounting standards, as it can with regular securities. In fact, the main framework for the regulation of capital markets in the U.S. (the